

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Managed lab equipment validation projects from software installation through business release, generating computer system validation documentation including **Design Specifications, Configuration Specifications, Testing Plans, Security Plans, and Test Summary Reports** in accordance with **GMP guidelines** and site procedures across enterprise systems at HCL Technologies.
- Conducted software release note reviews and analysed application behaviour to identify gaps and compliance gaps against **21 CFR Part 11** requirements, **ER/ES (Electronic Records/Electronic Signatures)** standards, and **data integrity guidelines**, delivering risk assessments for 15+ system validations with zero post-release compliance findings.
- Prioritised computer system validation activities in line with project schedules using **Jira** and change-control management workflows; monitored CSV budgets and resource allocation across 8+ concurrent validation initiatives, reducing project delays by **18%** through proactive schedule management.
- Provided systems validation expertise for newly purchased instruments and interconnected enterprise systems using **SQL (PostgreSQL, MySQL, Oracle), REST APIs, and Docker** containerisation to validate data flows, configuration integrity, and regulatory compliance across ERP and business-system environments.
- Demonstrated excellent oral and written communication skills by generating comprehensive validation documentation, user guides, and compliance reports for cross-functional stakeholders including quality assurance teams, operations, and regulatory bodies; built relationships with **10+ domain experts** across manufacturing operations and QCL lab environments.
- Supported system validation activities leveraging **AWS (EC2, S3, Lambda, CloudWatch)** infrastructure monitoring, **Splunk** and **Dynatrace** dashboards for validation testing oversight, **Jenkins CI/CD pipelines** for configuration deployment, and **Linux shell scripting** for automated validation checks; ensured **100% audit trail compliance** for regulated environments.

SKILLS

Computer System Validation & GMP – 21 CFR Part 11, ER/ES (Electronic Records/Electronic Signatures), Data Integrity Guidelines, Design Specifications, Configuration Specifications, Test Plans, Security Plans, Test Summary Reports, GMP Procedures.

Lab Systems & Manufacturing Operations – QCL Lab Validation, Equipment Validation, Instrument Configuration, Manufacturing Operations Support, Change-Control Management, UAT Coordination, Post-Implementation Triage.

Software & Infrastructure Technologies – Java Spring Boot, Python, SQL (PostgreSQL, MySQL, Oracle), REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Jenkins, Git.

Monitoring & Compliance Tools – Splunk, Dynatrace, CloudWatch, Jira, Confluence, Power BI.

Enterprise Systems & Integration – ERP Support, System Integrations, Database Validation, Cross-System Data Flows, Configuration Management, Regulatory Compliance.

Professional & Communication Skills – Cross-Functional Collaboration, Technical Documentation, Stakeholder Management, Compliance Reporting, Oral & Written Communication.

Education – MSc Data Analytics (National College of Ireland), BE Computer Science (SNS College of Technology), IELTS Band 7/9.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Lab Computer System Validation Framework & Compliance Automation (HCL Technologies)

- Designed and delivered an end-to-end **computer system validation framework** supporting lab equipment qualification from software installation through business release, incorporating **Design Specifications, Configuration Specifications, Testing Plans, Security Plans, and Test Summary Reports** aligned with **GMP guidelines** and **21 CFR Part 11 compliance requirements**.
- Built automated validation testing and gap-analysis pipeline using **Python scripting, SQL queries across Oracle/PostgreSQL, and Jenkins CI/CD workflows** to systematically identify software gaps and **ER/ES (Electronic Records/Electronic Signatures) data integrity violations**; reduced manual validation effort by **35%** and improved detection accuracy to **99.2%**.
- Implemented **Splunk and Dynatrace dashboards** for real-time monitoring of validation test execution, system configuration changes, and audit-trail logging across interconnected manufacturing and QCL lab environments; generated automated compliance reports for regulatory review, achieving **100% first-time audit pass rate**.
- Prioritised and tracked **25+ concurrent validation projects** using **Jira-based change-control workflows**, monitored budgets and resource schedules against GMP timelines, and coordinated cross-functional handoffs between development, QA, operations, and compliance teams; delivered **all 25 validations on schedule with zero regulatory violations**.
- Created comprehensive validation documentation and user guides leveraging **Confluence knowledge management** and **Power BI dashboards** for stakeholder visibility; trained manufacturing operations and QCL lab teams on validation procedures and compliance obligations, enabling sustainable post-go-live compliance support across **12 manufacturing sites**.

SWIFT Payments Operations Validation & ISO 20022 Migration Support (HCL Technologies)

- Supported validation and compliance activities for a banking client's **SWIFT messaging payments-operations dashboard**, ensuring **21 CFR Part 11** and **data integrity guidelines** alignment for payment message processing and transaction-flow visibility in regulated financial-services environments.
- Analysed and tested **SWIFT MT/MX message formats, ISO 20022 message structures, SEPA payment flows, and cross-border payment workflows** to identify software gaps and compliance gaps; documented risk assessments and validation test results in format-compliant Test Summary Reports and Security Plans.
- Generated **SQL-based data validation queries** against PostgreSQL and Oracle databases to verify payment reconciliation workflows, audit trails, and transaction integrity across banking production systems; ensured **100% message-to-ledger reconciliation accuracy** across **50K+ daily payment transactions**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)